ME46060 Engineering Optimizations

Exercise 0

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Course Name	Engineering Optimization	
Student Names	Marcos Carrera Valle	5682525
	Nicolás Nogueiras García	5683408
Responsible Instructor	Prof.dr.ir. M. Langelaar	
	Dr.ir. L.F.P. Noel	
	Dr. O. Nejadseyfi	
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Quantitative approach for coordination, at scale, of signalized intersection pairs

Jack Haddad, Nitzan Tur, Danny Veikherman, Eliav Buchnik, Shai Ferster, Tom Kalvari, Dan Karliner, Omer Litov, Avishai Zagoury, Ori Rottenstreich, Dotan Emanuel, and Avinatan Hassidim

Report Link: `https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10591202&casa_token=M26_evoDGVcAAAAA:L0ziJRxnEJv-5HC1EVKHxX2DKfKMXi_Kztf5lmuKMNkeLe2DgTcUQXyvH5mPjTghe8tYePpD&tag=1`

Project Green Light Link: `https://sites.research.google/gr/greenlight/`

1) Problem Considered

Project Green Light optimizes intersection traffic flow using AI and Google Maps driving trends. Instead of relying on physical sensors, it models traffic patterns to generate recommendations for existing traffic lights. The primary goal is to minimize greenhouse gases and maximize efficiency by reducing "stop-and-go" traffic, which is when a vehicle stops at a red light and needs to accelerate back again at the green light [1].

Within this context, the report by Haddad, J. et al. focuses on coordinating pairs of intersections to create "Green Waves" instead of optimizing a whole city's grid at once which is computationally too expensive. These "Green Waves" calculate the perfect timing to let cars pass through both intersections without stopping.

2) Optimization Parameters

The algorithm aims to minimize a weighted cost function of two parameters:

- Travel time t_{travel}.
- Fuel consumption f_{c}.

The trajectories of the vehicles traveling through the intersections can be used to estimate the average travel times and fuel consumptions. The latter is not as straight forward to compute as the first objective variable, and hence a model[2] is used.
The constraints imposed are the following:

- Hardware Independence: The solution must not require new sensors or communication hardware, relying solely on GPS data.
- Common Cycle Length: To coordinate both intersections, they must operate on the switch their lights at the same rate.

[1] stops can increase pollution by 29 times compared to open roads
[2] The model used is the National Renewable Energy Laboratory's Fast Simulation NREAL FASTsim

Within this context, the independent variable is the offset Δt. This is the time difference between the start of the green light at the first and second intersection. This is calculated from the (first intersection) crossing and (second intersection) entering times t_{cross} and t_{enter} respectively.

The cycle time t_{cycle}, which is he duration of the traffic light pattern for both intersections, is not considered an independent variable. The problem defined consists of intersections with already similar cycle lengths (maximum difference of 10%), hence the method simply sets both intersections to a common, average value.

3) Optimized Result

The method was analyzed using real data from Jakarta, Rio de Janeiro, Kolkata and Haifa. The method was then validated with a field experiment in Jakarta.

In the Jakarta example, intersections A (Jl. Kebon Sirih road with Jl. M. H. Thamrin road) and B (Jl. Kebon Sirih road and Jl. H. Agus Salim road) were considered (see fig. 1). The effect of offset is presented in figs. 2 and 3. It was seen that optimizing the value of t_{offset} for travel time (from 100 to 10 sec) could decrease travel time by 50% The optimal offset for fuel consumption was at $t_{\text{offset}} = 16$ sec, also reduced fuel consumption by approximately 50%. Hence the optimal general offset must be in the range of $10 - 16$ sec. With respect to the orginal offset, the travel time improvement and excessive fuel improvement were 15% and 31% respectively, or -5.63 sec and -1.8 mL

Figure 1: Intersections at Jl. Kebon Sirih road

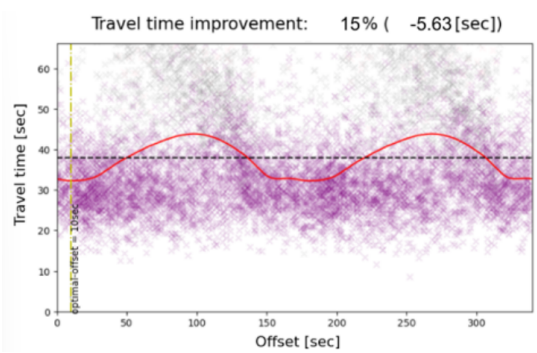

Figure 2: Offset effect on travel time

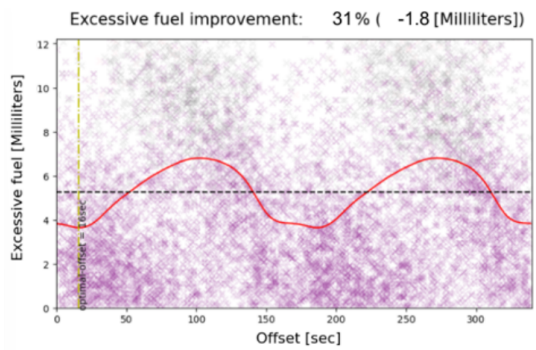

Figure 3: Offset effect on fuel consumption

Moreover, the validation case in the same intersection yielded an showed improvements of the objective variables of $\Delta t_{\text{travel}} = 5.1$ sec and $\Delta f_{\text{c}} = 1.7$ mL, which show the accuracy of the method when compared to the model predictions from figs. 2 and 3.